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A. Mathematical details

A.1. GHMMs

We consider non-Markovian sequence data generated by finite-dimensional generalized hidden Markov models (GHMMs)—a general class of models with latent transition operators[5] that includes as special cases (i) hidden Markov models (HMMs), and (ii) quantum hidden Markov models (QHMMs) (Riechers & Elliott, 2025). The latter describes repeated evolution and interaction with quantum density matrices, which is arguably the most general type of data-generating mechanism possible.[6]

A GHMM is defined by the four-tuple

$$\boldsymbol{\mathcal{M}} = \big(\mathcal{X}, \boldsymbol{\mathcal{S}}, \boldsymbol{\eta}^{(\varnothing)}, (T^{(x)})_{x\in\mathcal{X}}\big), \tag{7}$$

where $\mathcal{X}$ is the token alphabet, $\boldsymbol{\mathcal{S}} = \mathbb{R}^d$ is a real vector space that serves as the latent space of the GHMM, $\boldsymbol{\eta}^{(\varnothing)}$ is the initial row vector in $\boldsymbol{\mathcal{S}}$, and the net transition operator $T = \sum_{x\in\mathcal{X}} T^{(x)}$ has an eigenvalue of unity with associated right eigenvector $\mathbf{1} = T\mathbf{1}$, normalized such that $\boldsymbol{\eta}^{(\varnothing)}\mathbf{1} = 1$. The associated left eigenvector $\boldsymbol{\pi} = \boldsymbol{\pi}T$, also normalized such that $\boldsymbol{\pi}\mathbf{1} = 1$, is the *steady-state* of the GHMM. The ground-truth probability of any sequence $x_{1:\ell} \in \mathcal{X}^\ell$ of arbitrary length ℓ can be calculated directly via linear algebra:

$$Q_{\boldsymbol{\mathcal{M}}}(X_{1:\ell} = x_{1:\ell}) = \boldsymbol{\eta}^{(\varnothing)} T^{(x_{1:\ell})}\mathbf{1}\ , \tag{8}$$

where $T^{(x_{1:\ell})}$ is simply obtained by matrix multiplication $T^{(x_{1:\ell})} = T^{(x_1)}\cdots T^{(x_\ell)}$.

HMMs are a special case of GHMMs. The matrix elements of an HMM correspond to joint conditional transition probabilities $T^{(x)}_{s,s'} = Q(s', x|s)$ of observing token x and moving to state s' when starting in the hidden state s. In this special case, $\boldsymbol{\eta}^{(\varnothing)}$ is a probability distribution over the hidden states, and $\mathbf{1}$ is the column vector of all ones.

A.2. Belief updating implies geometry

The conditional probability density over the future $\overrightarrow{X}$ of any length, induced by some context $\overleftarrow{x}$, can be written in terms of the linear algebra of the GHMM as

$$Q(\overrightarrow{X}|\overleftarrow{x}) = \frac{\boldsymbol{\eta}^{(\varnothing)} T^{(\overleftarrow{x})}}{\boldsymbol{\eta}^{(\varnothing)} T^{(\overleftarrow{x})}\mathbf{1}} T^{(\overrightarrow{X})}\mathbf{1} = \boldsymbol{\eta}^{(\overleftarrow{x})} T^{(\overrightarrow{X})}\mathbf{1}\ . \tag{9}$$

This immediately highlights the important role of the *predictive vector*

$$\boldsymbol{\eta}^{(\overleftarrow{x})} := \frac{\boldsymbol{\eta}^{(\varnothing)} T^{(\overleftarrow{x})}}{\boldsymbol{\eta}^{(\varnothing)} T^{(\overleftarrow{x})}\mathbf{1}}\ , \tag{10}$$

and implies that, whenever two contexts $\overleftarrow{x}$ and $\overleftarrow{x}'$ induce the same predictive vector $\boldsymbol{\eta}^{(\overleftarrow{x})} = \boldsymbol{\eta}^{(\overleftarrow{x}')}$, these two contexts cannot affect the future differently. More generally, when two contexts $\overleftarrow{x}$ and $\overleftarrow{x}'$ induce nearby predictive vectors, these two contexts must make similar predictions about the future.

Altogether, the collection of these predictive vectors creates the *belief geometry* associated with the correlational structure of the sequence data. This belief geometry emerges in the activations of neural networks as they are pretrained on next-token prediction (Shai et al., 2024; Piotrowski et al., 2025; Riechers et al., 2025); during inference, the expected predictive vectors are induced as the model moves through context. Nonorthogonality of these vectors implies similar predictions induced by these contexts in the overlapping directions they share in activation space.

[5]The linearity of latent transition operators may seem limiting, yet these GHMMs can describe observations of paradigmatically nonlinear complex systems. There are two ways of seeing that the linearity in the latent space is sufficient to describe all systems: 1) Probability densities always map linearly according to the Ruelle–Perron–Frobenius operator, even for chaotic systems like the beloved Lorenz system. 2) Observations always map linearly according to the Koopman operator. These views are dual, in the linear algebraic sense.

[6]Although see Fanizza et al. (2024) and Riechers et al. (2025), which point out that generalized probabilistic theories (Plávala, 2023) allow for more compact representations, which are in fact still GHMMs.

A.3. Joint representation and the product-state submanifold

For a system composed of N parts—with the nth part having a local d_n-dimensional latent space $\mathbb{R}^{d_n}$—the most general predictive vector lives in a $((\prod_{n=1}^{N} d_n) - 1)$-dimensional affine subspace of $\mathbb{R}^{\prod_{n=1}^{N} d_n}$. We can also refer to such a general predictive vector as a *joint state*. For a classical system with a fixed interpretable orthonormal basis for the latent space, each predictive vector corresponds to a joint probability distribution over the composite latent states of all parts, and lives in the $((\prod_{n=1}^{N} d_n) - 1)$-dimensional probability simplex over joint latent states.

In generalized probabilistic theories like quantum mechanics (Plávala, 2023), it is always possible to choose a real vector space to represent the predictive states (e.g., generalized Bloch coordinates to represent the density matrix). In this case, there may not be a distinguished latent basis, but the predictive vectors will nevertheless all live in a convex space. For a single quantum system, this could be the Bloch ball for a qubit, or the generalized Bloch ball for larger quantum systems. These nominally post-classical systems allow for rather low-dimensional representations of the correlational structure of token sequences (Fanizza et al., 2024), and it has been shown that transformer use the real vector space of their residual stream to take advantage of these compressed representations (Riechers et al., 2025).

Following the terminology of multipartite quantum mechanics, a **product state** is a joint state that is equal to the tensor product of its reduced marginals. I.e., product states are of the form $\boldsymbol{\eta} = \bigotimes_{n=1}^{N} \boldsymbol{\eta}_n$. The set of all product states forms the **product-state manifold**—a curved low-dimensional manifold in the joint space. Since each reduced state $\boldsymbol{\eta}_n$ has only $d_n - 1$ degrees of freedom, the product-state manifold has an intrinsic dimension of only $\sum_{n=1}^{N}(d_n - 1)$ dimensions, although its linear span is the full $(\prod_{n=1}^{N} d_n)$-dimensional joint space.

Joint states that are off the product-state manifold encode correlations among parts. Sec. A.5 shows both (i) how to marginalize over other parts of the joint to obtain the *reduced state* of a subsystem, and (ii) how to quantify the *total correlation* among parts. In general, projecting a state onto the product-state manifold—by taking the tensor product of its reduced states—throws away information about how the parts are correlated, and thus compromises forecasts about the future. However, sometimes parts really are independent, or observations render the parts conditionally independent.

Recall that a GHMM generator is *conditionally independent* if each token-labeled linear-operator can be expressed as the tensor product of multiple linear operator factors: $T^{(x)} = \bigotimes_{n=1}^{N} T_n^{(x)}$. These conditionally independent operators map the product-state manifold to itself since $\left(\bigotimes_{n=1}^{N} \boldsymbol{\eta}_n\right)\left(\bigotimes_{n=1}^{N} T_n^{(x)}\right) = \bigotimes_{n=1}^{N} \boldsymbol{\eta}_n T_n^{(x)}$.

Accordingly, when (i) the initial predictive vector is a product state $\boldsymbol{\eta}^{(\varnothing)} = \bigotimes_{n=1}^{N} \boldsymbol{\eta}_n^{(\varnothing)}$ and (ii) the token-labeled latent transition operators are all of the same product form $T^{(x)} = \bigotimes_{n=1}^{N} T_n^{(x)}$, the belief updates maintain a tensor-product structure of the predictive vector[7] such that

$$\boldsymbol{\eta}^{(x_{1:\ell})} = \bigotimes_{n=1}^{N} \boldsymbol{\eta}_n^{(x_{1:\ell})} \overset{x_{\ell+1}}{\mapsto} \boldsymbol{\eta}^{(x_{1:\ell+1})} = \bigotimes_{n=1}^{N} \boldsymbol{\eta}_n^{(x_{1:\ell+1})} \tag{11}$$

for all ℓ and all $x_{1:\ell+1} \in \mathcal{X}^{\ell+1}$, then optimal forecasts about the future can be fully described via the collection of beliefs over each factor locally.[8] In this case, we say that the factors are *conditionally independent* given the context—although they are notably *not* necessarily independent if we marginalized over historical context.

These predictive vectors in Eq. (11) lie on the curved product-state manifold of $\sum_{n=1}^{N}\big(\dim(\boldsymbol{\mathcal{S}}_n) - 1\big)$ dimensions[9] within the larger tensor product space.[10] Crucially, the dimension of this submanifold grows only linearly in the number of parts, although the dimension of the tensor product space grows exponentially in the number of parts.

[7] in a fixed tensor-product decomposition of the vector space

[8] Even if the initial predictive vector is not a product state, the predictive vector will generically approach the product-state submanifold upon repeated application of product-preserving maps.

[9] The "-1" here comes from the fact that each constituent predictive vector $\boldsymbol{\eta}_n$ lives in an affine hyperplane of dimension $\dim(\boldsymbol{\mathcal{S}}_n) - 1$, since differences in these local predictive vectors are orthogonal to $\mathbf{1}_n = \left(\sum_{z^{(n)}} T_n^{(z^{(n)}|z^{(1:n-1)})}\right)\mathbf{1}_n$. If the sub-process is a (possibly input-dependent) HMM, then these local predictive vectors are belief states that live in the $\big(\dim(\boldsymbol{\mathcal{S}}_n) - 1\big)$-simplex over the HMM's hidden states.

[10] This is essentially the Segre variety from projective geometry.